SYSTEM CAPABILITIES BRIEF

NEPA 3D Knowledge Graph Review System

The current system is a local, auditable reviewer-engine for USDA Forest Service Region 1 NEPA source material. It turns workbook-governed source records into extracted evidence, authority-family models, source-claim links, applicability decisions, compliance outputs, decision-support reports, and a 3D knowledge graph surface.

Current operating status from local catalog, source-set graph, phase-eval, and promotion gates.

190 workbook source records in the local catalog	**35** authority families mapped to source records	**62/62** source-set graph checks	**26/26** current promotion gates

System process: controlled evidence to review support

The review engine keeps source capture, evidence, authority logic, and decision-support outputs traceable.

Workbook source contract	→	Audited local corpus	→	Authority graph	→	Review outputs
190		**160**		**35**		**26/26**
catalog records; row identity, scope, URLs		unique artifacts; hashes, manifests, SQLite		authority families; currentness, partitions,		current gates; matrix, support, QA, viewer

Design principle

Domain knowledge stays in data and artifacts: source rows, authority-family configs, rule templates, eval fixtures, graph contracts, and reviewer-facing reports.

What the system does

Maintains source identity, artifact provenance, citation labels, currentness status, extraction spans, graph edges, validation results, and reviewer-facing outputs as separate auditable surfaces.

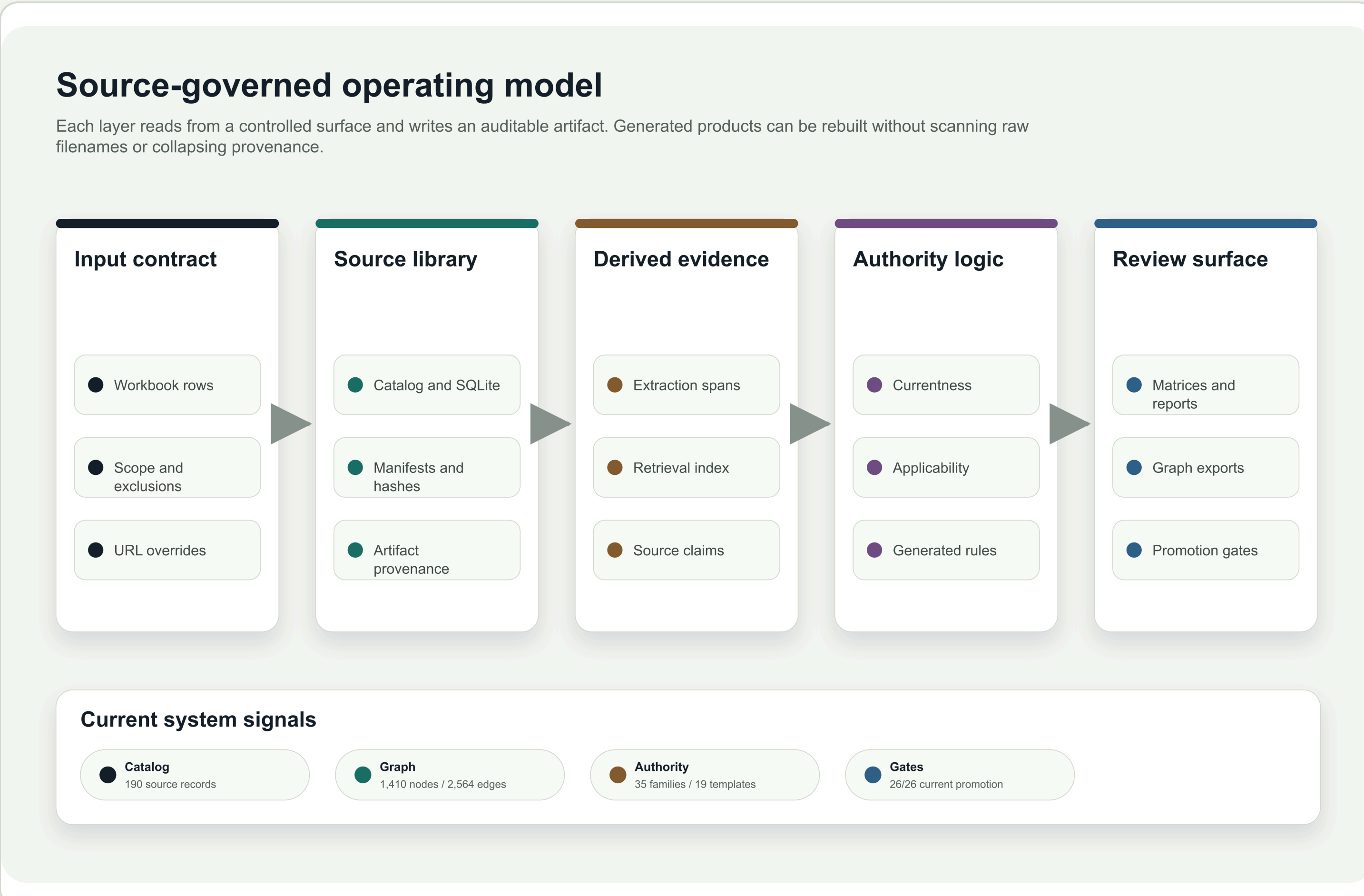
What the system avoids

It does not rely on hidden one-off shortcuts or unsupported legal conclusions. Expansion to new packages or profiles is routed through source, profile, and evaluation gates.

CAPABILITY 1 / SOURCE-GOVERNED OPERATING MODEL

Controlled source library to graph-ready review layer

The workbook remains the contract. Downloader, catalog, extraction, retrieval, evidence graph, source-claim graph, rule binding, and knowledge graph exports all preserve row identity and provenance so downstream review products can be replayed and inspected.



Operating model: source records are captured once, then promoted through derived layers with explicit validation. The graphic separates source capture, semantic extraction, applicability, review products, and visualization so the same evidence model supports multiple outputs without duplicating logic.

Currentness and partitioning

Active review sources are separated from candidate, blocked, superseded, and currentness-only records before graph export. That prevents stale or excluded sources from silently supporting current authority.

Validation before display

Graph exports carry node and edge type checks, provenance requirements, endpoint compatibility checks, readiness status, and failure-category counts before they are used by the viewer or promotion suite.

Current graph surface: 1,410 nodes and 2,564 edges.

Catalog boundary: 160 unique artifacts linked to workbook records.

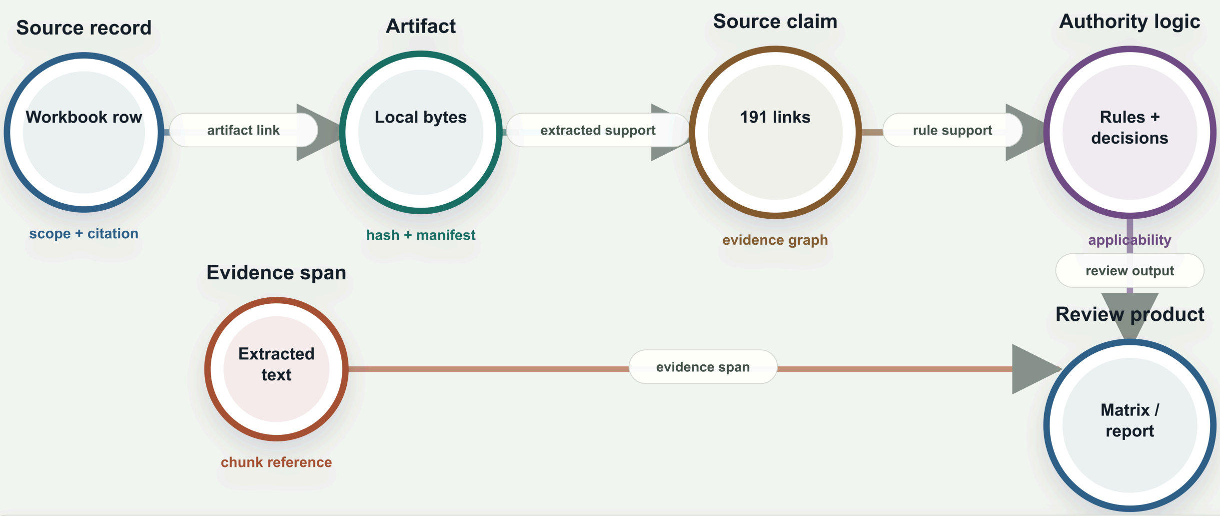
CAPABILITY 2 / EVIDENCE TRACEABILITY AND REVIEW PRODUCTS

Every finding stays connected to evidence

The system keeps source records, artifacts, chunks, evidence spans, source claims, rule templates, applicability decisions, generated rules, and findings as graph-visible objects. Review outputs are produced from those artifacts rather than from a narrative-only analysis pass.

Evidence path stays inspectable

The system treats a finding as the end of a traceable path, not as a standalone sentence. Each link has a provenance role and can be tested.



Review outputs from the trace

- compliance matrix
- decision-support report
- final QA packet
- knowledge graph export
- 3D viewer scenes

Traceability model: source evidence supports claims; claims support rules and applicability; applicability drives review products. The chain remains inspectable across JSON, Markdown, PDF, SQLite, and graph export surfaces.

Reviewer-facing outputs

- Compliance matrix in JSON, Markdown, and PDF.
- Decision-support report with evidence paths and residual risk.
- Final QA certification and replay validation artifacts.
- Knowledge graph JSON, node/edge files, summaries, and validation.

Reverse compliance

The same evidence model can search for unsupported claims, older-regulation dependencies, missing source support, unresolved applicability, and forest-plan component gaps before the package is treated as reviewer-ready.

CAPABILITY 3 / READINESS, GOVERNANCE, AND EXPANSION

Promotion gates make readiness explicit

Readiness is not inferred from a finished-looking report. The system uses deterministic gates for source capture, currentness, extraction, retrieval, graph validation, applicability, compliance outputs, forest-plan evaluation, decision support, final QA, and promotion.

Readiness is a gated state

Reports, PDFs, and viewer scenes are downstream products. The system first verifies source capture, evidence, graph validity, review outputs, and promotion status.

Current gate stack

Catalog integrity	Workbook coverage, source statuses, artifact links	PASS
Currentness	Active, candidate, superseded, and blocked source roles	PASS
Graph validation	62/62 source-set checks	PASS
Phase evaluation	20/20 review phases	PASS
Promotion suite	26/26 current gates	PASS

Expansion boundary

Forest-plan profiles	1/10	graph-ready
Blocked profiles	9	visible readiness blockers
Components	329	inventory-backed
Profile requirements	8	directives + overlays

Blocked or incomplete sources remain explicit so expansion risk is visible before a package is called ready.

Source control
Workbook rows, URL overrides, catalog records, hashes, and source-set manifests.

Evidence build
Extraction, retrieval, evidence spans, graph edges, and source-claim links.

Authority model
Authority families, currentness decisions, rule templates, and source partitions.

Applicability
Package-specific decisions, generated rules, screened-out authority, and unresolved items.

Review outputs
Compliance matrix, decision support, final QA, and reviewer-facing PDFs.

Viewer surface
3D graph scenes driven by validated graph JSON rather than hard-coded visuals.

Expansion boundary

New packages, forest profiles, or authority deltas are added by expanding the source/profile contract and replaying gates. Blockers remain visible when profiles or sources are not ready, instead of being collapsed into a generic failure.

Boundary statement: the system produces auditable review support and readiness evidence. It does not replace professional judgment, agency review, or legal sufficiency review.